

HTML Basics: Theory

Lecture 1 — Fundamentals of Web Design & Development (CT053-3-1)

1. HTML vs. XHTML

XHTML is a stricter, XML-based version of HTML. Modern HTML5 is tolerant of markup from all earlier versions, but it's worth knowing the stricter rules XHTML introduced — good habits either way:

- Tags and attribute names must be lowercase.
- Every tag must be closed — `<br />`, `<img />` instead of `<br>`, `<img>`. HTML allows unclosed tags like `<p>par1 <p>par2`, XHTML does not.
- Only one root `<html>` element is allowed (HTML tolerates more than one).
- Attribute minimization is forbidden: write `checked="checked"`, not just `checked`.

Takeaway: browsers tolerate messy HTML and silently fix errors — you shouldn't rely on that. Valid, well-formed code loads faster and is easier to debug.

2. Elements, Tags & Attributes

An **element** is a piece of structure on the page. It's written using a pair of **tags**: an opening tag and a closing tag, with content in between.

```
<p>This is a paragraph.</p>
```

Some elements are **self-closing** (no content, no separate closing tag) — e.g. `<br />`, `<img />`.

Tags can carry **attributes**: `name="value"` pairs inside the opening tag that configure the element or add information.

```
<a href="http://www.apu.edu.my/" title="APU Website">APU</a>
```

Here **href** and **title** are attributes. A few attributes apply to almost any element: **id** (unique per page), **class**, **style**, **title** (shown as a tooltip on hover).

Elements can be **nested** inside one another — this nesting is what gives the page its structure.

3. Overall HTML Document Format

Every HTML page follows the same skeleton:

```
<!DOCTYPE HTML>
<html>
<head>
<title>My First HTML Page</title>
</head>
<body>
<p>This is some text...</p>
</body>
</html>
```

- **<!DOCTYPE HTML>** — first line of the file; tells the browser which version of HTML the page uses.
- **<html>...</html>** — the single root element wrapping the whole page.
- **<head>...</head>** — page information not shown directly on screen (must contain exactly one `<title>`; can also hold `<meta>`, `<style>`, `<script>`).
- **<body>...</body>** — the visible content of the page.

Formatting convention: start every block element on a new line and indent nested elements. Browsers ignore extra whitespace, so this costs nothing and makes debugging far easier.

4. Key Tags: Headings, Links & Images

Tag	Purpose	Example
<code><h1> ... <h6></code>	Headings, largest to smallest	<code><h1>Title</h1></code>
<code><p></code>	Paragraph of text	<code><p>Some text</p></code>
<code>
</code>	Line break	Line 1 Line 2
<code></code>	Hyperlink (anchor tag)	<code>Link</code>
<code></code>	Image	<code></code>
<code></code> , <code></code>	Bold emphasis, italic emphasis	<code>bold</code>
<code><div></code> , <code></code>	Generic block / inline containers	<code><div class="box">...</div></code>
<code><!-- comment --></code>	Comment (ignored by browser)	<code><!-- note --></code>

5. The Anchor Tag `<a>` in Detail

The anchor tag creates a hyperlink. Its key attribute is **href** (hyperlink reference) — the destination URL:

```
<a href="http://www.telerik.com/" title="Telerik">Link to Telerik</a>
```

href: destination address (a full URL, a relative path, or #section on the same page).

title: tooltip text shown on hover.

The text between the opening and closing tag is what the user sees and clicks.

6. The Image Tag `<img>` in Detail

The image tag embeds a picture. It is self-closing and has two important attributes:

```

```

src (required): path or URL to the image file.

alt (required): alternate text shown if the image fails to load, and read by screen readers.

Next: hands-on tutorial — building and viewing your first HTML page.